Bayesian inference

by Marco Taboga, PhD

Bayesian inference is a way of making statistical inferences in which the statistician assigns subjective probabilities to the distributions that could generate the data. These subjective probabilities form the so-called prior distribution.

After the data is observed, Bayes' rule is used to update the prior, that is, to revise the probabilities assigned to the possible data generating distributions. These revised probabilities form the so-called posterior distribution.

This lecture provides an introduction to Bayesian inference and discusses a simple example of inference about the mean of a normal distribution.

Review of the basics of statistical inference

Remember the main elements of a statistical inference problem:

1. we observe some data (a sample), that we collect in a vector ;
2. we regard as the realization of a random vector ;
3. we do not know the probability distribution of (i.e., the distribution that generated our sample);
4. we define a statistical model, that is, a set of probability distributions that could have generated the data;
5. optionally, we parametrize the model, that is, we put the elements of in correspondence with a set of real vectors called parameters;
6. we use the sample and the statistical model to make a statement (an inference) about the unknown data generating distribution (or about the parameter that corresponds to it).

In Bayesian inference, we assign a subjective distribution to the elements of , and then we use the data to derive a posterior distribution.

In parametric Bayesian inference, the subjective distribution is assigned to the parameters that are put into correspondence with the elements of .

The likelihood

The first building block of a parametric Bayesian model is the likelihood

The likelihood is equal to the probability density of when the parameter of the data generating distribution is equal to .

For the time being, we assume that and are continuous. Later, we will discuss how to relax this assumption.

Example

Suppose that the sample

is a vector of independent and identically distributed draws from a normal distribution.

The mean of the distribution is unknown, while its variance is known. These are the two parameters of the model.

The probability density function of a generic draw is

$$p(x_i|\mu) = \left(2\pi\sigma^2\right)^{-1/2} \exp\left(-\frac{1}{2}\frac{(x_i-\mu)^2}{\sigma^2}\right)$$

where we use the notation to highlight the fact that is unknown and the density of depends on this unknown parameter.

Because the observations are independent, we can write the likelihood as

$$p(x|\mu) = \prod_{i=1}^{n} p(x_i|\mu) = \left(2\pi\sigma^2\right)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

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The prior

The second building block of a Bayesian model is the prior

The prior is the subjective probability density assigned to the parameter .

Example

Let us continue the previous example.

The statistician believes that the parameter is most likely equal to and that values of very far from are quite unlikely.

She expresses this belief about the parameter by assigning to it a normal distribution with mean and variance .

So, the prior is

$$p(\mu) = \left(2\pi\tau^2\right)^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu-\mu_0)^2}{\tau^2}\right)$$

The prior predictive distribution

After specifying the prior and the likelihood, we can derive the marginal density of :

where: in step we perform the so-called marginalization (see the lecture on random vectors); in step we use the fact that a joint density can be written as the product of a conditional and a marginal density (see the lecture on conditional probability distributions).

The notation

is a shorthand for the multiple integral

where is the dimension of the parameter vector .

The marginal density of , derived in the manner above, is called the **prior predictive distribution**. Roughly speaking, it is the probability distribution that we assign to the data before observing it.

Example

Given the prior and the posterior specified in the previous two examples, it can be proved that the prior predictive distribution is

$$p(x) = \int_{-\infty}^{\infty} p(x|\mu)p(\mu)d\mu$$
$$= (2\pi\sigma^2)^{-n/2}\left|\det\left(I_n + \frac{\tau^2}{\sigma^2}ii^\top\right)\right|\exp\left(-\frac{1}{2\sigma^2}(x - i\mu_0)^\top\left(I + \frac{\tau^2}{\sigma^2}ii^\top\right)^{-1}(x - i\mu_0)\right)$$

where is an vector of ones, and is the identity matrix.

Hence, the prior predictive distribution of is multivariate normal with mean and covariance matrix

$$\sigma^2\left(I + \frac{\tau^2}{\sigma^2}ii^\top\right) = \sigma^2 I + \tau^2 ii^\top$$

Thus, under the prior predictive distribution, a draw has mean , variance and covariance with the other draws equal to .

The covariance is induced by the fact that the mean parameter , which is stochastic, is the same for all draws.

The posterior

After observing the data , we use Bayes' rule to update the prior about the parameter :

The conditional density is called posterior distribution of the parameter.

By using the formula for the marginal density derived above, we obtain

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{\int_\theta p(x|\theta)p(\theta)d\theta}$$

Thus, the posterior depends on the two distributions specified by the statistician, the prior and the likelihood .

Example

In the normal model of the previous examples, it can be proved that the posterior is

$$p(\mu|x) = (2\pi\sigma_n^2)^{-1/2}\exp\left(-\frac{1}{2}\frac{(\mu - \mu_n)^2}{\sigma_n^2}\right)$$

where

$$\mu_n = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}\left[\frac{n}{\sigma^2}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) + \frac{1}{\tau^2}\mu_0\right]$$
$$\sigma_n^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}$$

Thus, the posterior distribution of is normal with mean and variance .

The posterior mean is a weighted average of:

- the mean of the observed data ( );
- the prior mean .

The weights are inversely proportional to the variances of the two means:

- if the prior variance is high, then the prior mean receives little weight;
- by the same token, if the variance of the sample mean (which is equal to) is high, then the sample mean receives little weight and more weight is assigned to the prior.

Both the sample mean and the prior mean provide information about . They are combined together, but more weight is given to the signal that has higher precision (smaller variance).

When the sample size becomes very large (goes to infinity), then all the weight is given to the information coming from the sample (the sample mean) and no weight is given to the prior. This is typical of Bayesian inference.

The posterior predictive distribution

Suppose that a new data sample is extracted after we have observed and we have computed the posterior distribution of the parameter

Assume that the distribution of depends on , but is independent of conditional on :

Then the distribution of given is

The distribution of given , derived in the manner above, is called the **posterior predictive distribution**.

Example

In the normal model of the previous examples, the prior is updated with draws .

Consider a new draw from the same normal distribution.

It can be proved that the posterior predictive distribution of is a normal distribution with mean (the posterior mean of) and variance , where is the posterior variance of .

Integrals

Up to know we have assumed that and are continuous. When they are discrete, there are no substantial changes, but probability density functions are replaced with probability mass functions and integrals are replaced with summations.

For example, if is discrete and is continuous:

- the marginal density of becomes

  where is the probability mass function of , and the summation is over all possible values of ;
- the formula for the posterior probability mass function of is the same as in the continuous case:

Proportionality

We now take a moment to explain some simple algebra that is extremely important in Bayesian inference.

Given a posterior density

we can take any function of the data that does not depend on , and we can use it to build another function

Since the data is considered a constant after being observed, we write

that is, is proportional to .

The posterior can be recovered from as follows:

where: in step we use the fact that does not depend on and, as a consequence, it can be brought out of the integral; in step we use the fact that the integral of a density (over the whole support) is equal to .

In summary, when we multiply the posterior by a function that does not depend on (but may depend on), we obtain a function proportional to the posterior.

If we divide the new function by its integral, then we recover the posterior.

The posterior is proportional to the prior times the likelihood

In the posterior formula

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{\int_\theta p(x|\theta)p(\theta)d\theta} \quad (1)$$

the marginal density

does not depend on (because is "integrated out").

Thus, by using the notation introduced in the previous section, we can write

that is, the posterior is proportional to the prior times the likelihood .

Both and are known because they are specified by the statistician.

Thus, the posterior (which we want to compute) is proportional to the product of two known quantities.

This proportionality to two known quantities is extremely important in Bayesian inference: various methods allow us to exploit it in order to compute the posterior when (2) cannot be calculated and hence (1) cannot be worked out directly.

Factorization

Often, we are not able to apply Bayes' rule

because we cannot derive the marginal distribution analytically.

However, we are sometimes able to write the joint distribution

as

where:

- is a function that depends only on ;
- is a probability density (or mass) function of (for any fixed).

If we can work out this factorization, then

See the lecture on the factorization of probability density functions for a proof of this fact.

MCMC

There are several Bayesian models that allow us to compute the posterior distribution of the parameters analytically. However, this is often not possible.

When an analytical solution is not available, Markov Chain Monte Carlo (MCMC) methods are commonly employed to derive the posterior distribution numerically.

MCMC methods are Monte Carlo methods that allow us to generate large samples of correlated draws from the posterior distribution of the parameter vector by simply using the proportionality

The empirical distribution of the generated sample can then be used to produce plug-in estimates of the quantities of interest.

See the lecture on MCMC methods for more details.

Quantities of interest

After updating the prior, we can use the posterior distribution of to make statements about the parameter or about quantities that depend on .

The quantities about which we make a statement are often called **quantities of interest** (e.g., Bernardo and Smith 2009) or **objects of interest** (e.g., Geweke 2005).

The Bayesian approach provides us with a posterior probability distribution of the quantity of interest. We are free to summarize that distribution in any way that we deem convenient.

For example, we can:

- plot the probability density (or mass) of the quantity of interest;
- report the mean of the distribution (as our best guess of the true value of the quantity of interest) and its standard deviation (as a measure of dispersion of our posterior beliefs);
- report the probability that the quantity of interest (say, a parameter) is equal (or very close) to a certain value which had previously been hypothesized (similarly to what is done in hypothesis testing).

More examples of Bayesian inference

Now that you know about the basics of Bayesian inference, you can study two applications in the following lectures:

- Bayesian inference about the parameters of a normal distribution, where we prove all the formulae shown in the examples above;
- Bayesian inference about the parameters of a linear regression model.

References

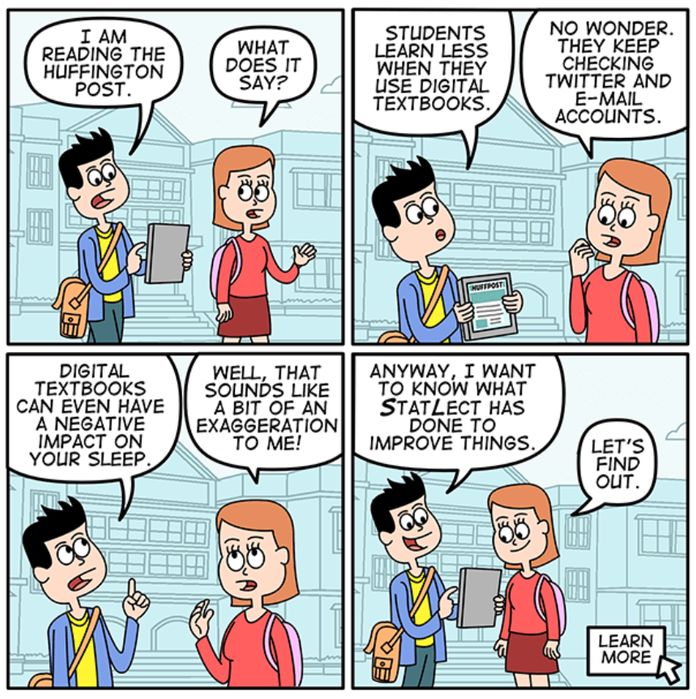
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